Sahaj Bhadja

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Summary

Software Engineer with 2+ years building secure backend systems and IEEE-published **cryptography** research on lattice-based post-quantum signatures, with **authentication**, access control, secure coding, design patterns, and rigorous **testing**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed multi-tenant storage on Azure SQL and Blob Storage with **row-level security** that enforces strict tenant data isolation for 500+ users, and tuned indexed queries to cut latency from 1.2s to 350ms.
- Built fault-tolerant job processing with explicit state transitions and **failure isolation**, handling 1K+ jobs at 3x throughput, and shipped validated integrations through CI/CD at 99%+ reliability across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed **secure file ingestion** for PDF, DOCX, and ZIP uploads with Flask-Login authenticated sessions, processing 1K+ untrusted files per batch for 150+ instructors and 10K+ daily requests.
- Debugged timeouts and failures across external model providers by adding fallback routing and timeout handling, keeping production APIs stable at peak load and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented modules with Singleton and Factory **design patterns** for cross-platform data-sharing software, covered by 20+ automated unit and integration suites in PyTest and Mockito run nightly.
- Troubleshot large-scale API integrations from logs and tests, improving response times 30% and cutting technical debt 20% while collaborating with Microsoft and GitHub stakeholders in Agile sprints.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a quantum-resistant blockchain integrating 5+ **lattice-based cryptographic** primitives, including NIST-selected **FALCON** signatures, with node-behavior monitoring and Borda Count ranking for fair witness selection in the network.
- Benchmarked the post-quantum design against an ECDSA baseline across 100 to 1,000 transaction workloads, sustaining 957.8 TPS at 224ms, and presented the work at IISc Bengaluru in an IEEE paper.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Implemented **wallet-signature authentication** for passwordless login, verifying signed challenges instead of storing secrets and eliminating password storage for 100% of wallet users across Android, iOS, and web clients.
- Built PCI-oriented payment flows with JWT-secured APIs and **idempotent webhooks** for refunds, disputes, and chargebacks, cutting manual payment handling about 70% across 50K+ booking and payment records in production.

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Solidity, Consensus Algorithms
Security & Crypto: Post-Quantum Cryptography, Lattice-Based Signatures, FALCON, ECDSA, Digital Signatures, Authentication
Secure Design: Access Control, Row-Level Security, Role-Based Access, Self-Sovereign Identity, Secure Coding, Smart Contracts
Engineering: Design Patterns, Object-Oriented Design, Unit Testing, PyTest, Mockito, Debugging, Code Reviews, Blockchain
Platforms: Linux, Git, Docker, CI/CD, AWS, Azure, PostgreSQL, REST APIs, Web3, Bash, JWT, Agile

Projects

Authenticated Encryption Library | *Rust, AES-GCM, XChaCha20-Poly1305, Argon2, HKDF*

- Designed a **Rust** library wrapping AEAD encryption with Argon2 key derivation and HKDF subkeys behind a misuse-resistant API, verified with official test vectors, fuzzing, and **constant-time** comparison checks.

Post-Quantum Key Exchange Prototype | *Go, ML-KEM, X25519, Go testing*

- Designed a **Go** prototype of hybrid key exchange combining X25519 with **ML-KEM**, following the published specification, validated against known-answer tests and benchmarked for handshake latency and key and ciphertext sizes.

Blockchain for User Authentication Paper | *IEEE INCOFT 2023, Smart Contracts, Self-Sovereign Identity*

- Co-authored an IEEE paper analyzing blockchain **authentication** for users, IoT devices, and role-based access, examining defenses against identity theft, replay, man-in-the-middle, and phishing attacks against centralized identity stores and verifiers.